PRICING OF FUTURES CONTRACTS AT INCEPTION

Learning Outcome

- compare the value and price of forward and futures contracts

When a forward commitment is initiated, no cash is exchanged and the contract is neither an asset nor a liability to the buyer or seller. The value of both a forward contract and a futures contract at initiation is zero:

$$V_0(T) = 0. \quad (1)$$

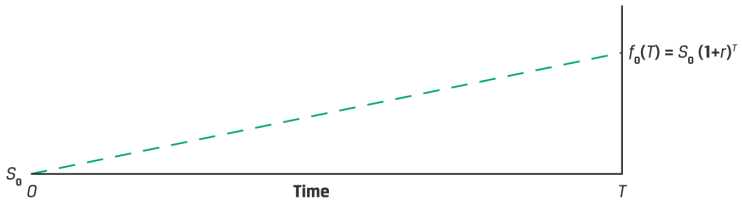
An underlying asset with no cost or benefit has a futures price $f_0(T)$ at $t = 0$ of:

$$f_0(T) = S_0(1 + r)^T, \quad (2)$$

where r is the risk-free rate and T is the time to maturity. As in the case of a forward contract, the futures price is the spot price compounded at the risk-free rate over the life of the contract. This is shown in Exhibit 1, where the slope of the line is equal to the risk-free rate, r.

Exhibit 1:

Futures Price at Initiation



▶ Expand / collapse Extended Description

As for forwards, we use discrete compounding as in Equation 2 for futures on individual underlying assets. However, for underlying assets that are comprised of a portfolio—such as an equity, fixed-income, commodity, or credit index—or where the underlying involves foreign exchange with interest rates denominated in two currencies, continuous compounding is the preferred method, as shown in Equation 3:

$$f_0(T) = S_0e^{rT}. \quad (3)$$

EXAMPLE 1

As shown in a previous lesson, Procam Investments purchases a 100-ounce gold futures contract. The current spot price is $1,770.00 per ounce, the risk-free rate is 2.0%, and we assume gold may be stored at no cost. Calculate the no arbitrage futures price, $f_0(T)$, for settlement in 91 days (T = 91/365 or 0.24932).

Hide Solution

Solution:

Using Equation 2: $f_0(T) = S_0(1 + r)^T$

$\$1{,}778.76$ per ounce $= \$1{,}770.00 \times (1.02)^{0.24932}$

Contract price = $177,876.04 (= 100 × $1,778.76)

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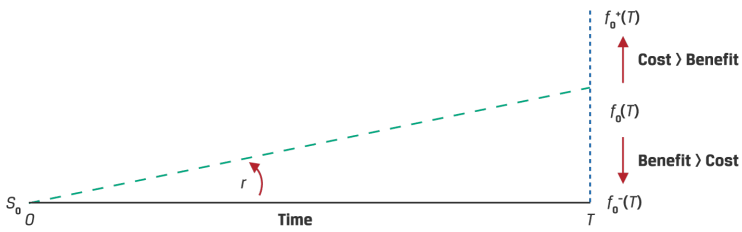
As in the case of a forward, for underlying assets with ownership benefits or income (I) or costs (C) expressed as a known amount in present value terms at time $t = 0$, the spot versus futures price relationship using discrete compounding is shown in Equation 4 and Exhibit 2:

$$f_0(T) = [S_0 - PV_0(I) + PV_0(C)]\,(1 + r)^T. \qquad 4$$

Exhibit 2:

Futures Prices with Underlying Asset Costs and Benefits



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EXAMPLE 2

Procam purchased a gold futures contract in Example 1 at $f_0(T)$ of \$177,876.04 (or \$1,778.76 per ounce) with S_0 equal to \$1,770 per ounce. How would $f_0(T)$ change to satisfy the no arbitrage condition if a \$2 per ounce cost of gold storage and insurance were payable at the end of the contract?

The futures price for a commodity with known storage cost amounts may be determined using Equation 4, where $PV_0(I) = 0$:

$$f_0(T) = [S_0 + PV_0(C)]\,(1 + r)^T.$$

First, solve for the present value of the storage cost per ounce, $PV_0(C)$, as follows:

$$PV_0(C) = \$1.99 = [\$2(1.02)^{0.24982}].$$

Substitute $PV_0(C) = \$1.99$ into Equation 4 to solve for $f_0(T)$:

$$f_0(T) = (\$1{,}770.00 + \$1.99)(1.02)^{0.24982}$$

$$= \$1{,}780.78 \text{ per ounce}$$

The addition of storage and insurance costs increases the difference between the spot price and futures price. Finally, as in the case of forwards, a futures price, $f_0(T)$, significantly *below* the no arbitrage price including cash costs and benefits may indicate the presence of a convenience yield.

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Discussion

What do you want to discuss... (required)

SH If Example 2 is following on from Example 1, then shouldn't T = 0.24932, not 0.24982?

YW example 2: PV0(C) = $1.99 = [$2(1.02)0.24982] should read PV0(C) = $1.99 = [$2/(1.02)0.24982].

Created 2 months ago by Yue Wang

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